

PHILIPPINE CARABAO CENTER · TRAINING COURSE · 13-19 SEPT 2026

Reproduction in Cattle & Water Buffalo

**From Reproductive Fundamentals to Artificial
Insemination,
Embryo Transfer & Smart-Farm Precision Breeding**

Basics

Estrus & AI

Embryo Transfer / IVF

Smart Farm

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25+ years bringing reproductive biotechnology from the lab to the farm

I am a veterinarian working across the full reproduction chain — clinical practice, a biotechnology laboratory, a genetics company, and a working dairy farm. Everything in this course is what we do every week in the field.



Goryeo VET Clinic

Field AI, estrus synchronization & herd reproduction consulting



L&H Institute of Biotechnology

OPU-IVF, MOET, embryo culture & cryopreservation



Genetics Co.

Genomic selection, sexed embryos & sire evaluation



Song Young Shin Farm

Sensor-based smart dairy — reproduction data every day

Reproduction is the **engine of genetic improvement** — the fastest, cheapest way to upgrade a national herd. That makes it the heart of the carabao development mission.

Five steps, one goal: more calves, better genetics

1

Foundations

Anatomy, the estrous cycle & the hormones that drive it

2

Estrus & heat detection

Reading heat — and the buffalo "silent heat" problem

3

Artificial insemination

Semen, timing, technique & conception rate

4

Embryo transfer & IVF

Multiplying elite female genetics fast

5

Smart farm

Sensors & data for precision reproduction

Cattle principles throughout, with water-buffalo (carabao) specifics highlighted in **rose**.

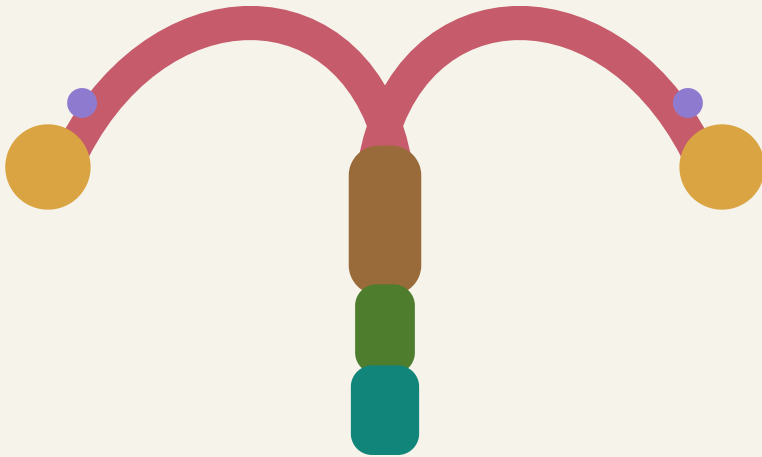
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PART ONE

Reproductive Foundations

Anatomy of the female tract · the estrous cycle · hormonal control · follicular dynamics · the male contribution. You cannot manage what you do not understand.

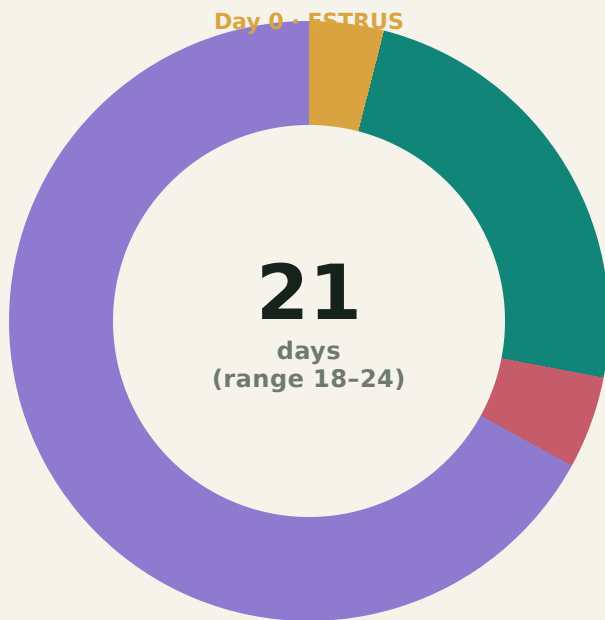
The female reproductive tract



- Ovary** — houses follicles & the corpus luteum; source of eggs & hormones
- Oviduct (fallopian tube)** — site of fertilization
- Uterine horns** — where the embryo implants & grows
- Body of uterus** — the AI semen deposition site
- Cervix** — the gateway we pass the AI gun through
- Vagina** — receives natural service

Rectovaginal ("recto-cervical") AI works by feeling the cervix through the rectum wall and guiding the gun by hand.

The estrous cycle — a ~21-day rhythm



Proestrus → Estrus (Day 0)

Follicle matures, estrogen peaks → **standing heat**. Ovulation ~24-30 h after heat onset.

Metestrus → Diestrus (Day 1-17)

Corpus luteum (CL) forms & secretes **progesterone** — maintains pregnancy or holds the cycle.

Luteolysis (Day 17-18)

If not pregnant, the uterus releases **PGF2α** → the CL regresses → the next cycle begins.



Cycle length similar (~21 d) but estrus is **shorter, weaker & often silent** — and strongly **seasonal**. We will return to this.

The hormonal chain of command

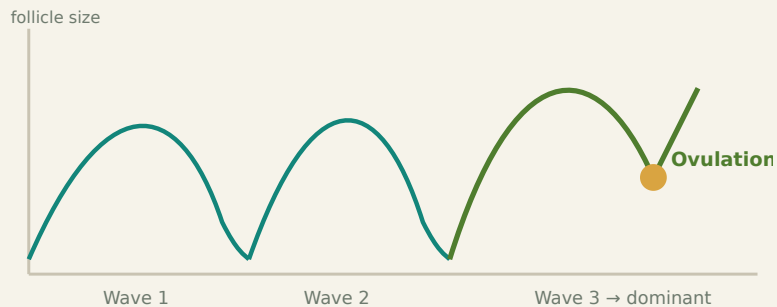
Every reproductive tool we use — synchronization, AI timing, superovulation — works by pushing one of these four levers.



PGF2 α (from the uterus) destroys the CL → this is the "reset button" we use to synchronize heat.

Negative feedback: high progesterone suppresses GnRH/LH — remove it, and the cow returns to heat.

Follicular waves & ovulation



- Follicles grow in **2-3 waves** per cycle (buffalo usually 1-2)
- Each wave has a **dominant follicle**; only the last one ovulates
- The **LH surge** triggers ovulation ~24-30 h after heat onset
- The egg is viable only **~6-12 h** — timing is everything

Why this matters:

Understanding waves is the basis of **synchronization & superovulation** — we manipulate which follicles grow and when they ovulate.

The sire side: one bull, thousands of daughters



Spermatogenesis

~60 days to make sperm in the testis. Heat stress & illness 2 months ago still affect today's fertility.



Semen evaluation

Volume · concentration · **motility** · morphology. Only quality ejaculates are processed.



The power of AI

One elite bull → **thousands of frozen straws**, shipped worldwide, used long after his lifetime.



Buffalo

River-type bulls (e.g. **Murrah**) are the key to upgrading native swamp buffalo — AI puts that superior sire in every village.

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PART TWO

Estrus & Heat Detection

The single biggest cause of AI failure worldwide is not bad semen — it is **missed heat**. For buffalo, this is the central challenge.

Signs of estrus (heat)

The one sure sign — standing to be mounted

She stands still when mounted by another animal. This is the reference point for AI timing.

Secondary signs

Mounting others

Clear mucus discharge

Restlessness / bawling

↓ feed & milk

Swollen vulva

Chin-resting / sniffing

The buffalo problem

- **Silent / weak heat** — few or no visible signs
- **Mostly at night** — missed by day-time checks
- **Seasonal** — poorer expression in hot months
- Result: **low AI conception** from missed heat

→ This is exactly the gap sensors & timed-AI close (Parts 3 & 5).

How we catch heat — low-tech to high-tech

Visual observation

3× daily, ≥20 min, quiet times (early AM / late PM). Free but labour-heavy & misses night heat.

Tail paint / chalk

Rubbed-off paint = she was mounted. Cheap, effective aid to observation.

Teaser / heat-mount detectors

Vasectomized teaser bull or pressure-activated patches (KaMaR).

Pedometers / activity

Heat ↑ walking 2–4×. Automated & objective — good for silent heat.

Rumen & body sensors

Temperature + rumination + activity → heat alerts, day or night. **Best fit for buffalo.**

Ultrasound / palpation

Confirms follicle/CL & pregnancy — the vet's cross-check.

Timing: the AM / PM rule

Sperm need ~**6 h** in the tract to become fertile (capacitation) and live ~24 h; the egg lives only ~**6-12 h**. So we inseminate **late in heat / just before ovulation**.

Heat seen in the **MORNING** → inseminate that **EVENING**

Heat seen in the **EVENING** → inseminate next **MORNING**

Rule of thumb: AI about 12 hours after first standing heat; ovulation follows ~24-30 h after heat onset.

6-12 h

egg viability window

~24 h

sperm survival in tract



Silent heat makes "onset" hard to judge → **timed AI** (Part 3) removes the guesswork.

Estrus synchronization — scheduling the herd

Instead of waiting for each animal, we bring a whole group into heat together — enabling planned, batch AI.

PGF2 α (single / double)

Destroys the CL → heat in 2-5 days. Simple & cheap; needs an active CL (cycling animals).

Progesterone device (CIDR / PRID)

Intravaginal insert mimics the CL, then withdrawn. **Works on non-cycling / silent buffalo** & anestrus.

Ovsynch (→ Fixed-time AI)

GnRH → PGF2 α → GnRH controls ovulation itself → AI by the clock, **no heat detection needed**.

 **Buffalo** Progesterone + Ovsynch-type protocols are the **practical answer to silent, seasonal heat** — we detail the timeline next.

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PART THREE

Artificial Insemination

The most powerful, lowest-cost genetic tool ever put in a farmer's hands
— and the backbone of any national herd-improvement program.

Why artificial insemination?



Rapid genetic gain

Access the world's proven sires — upgrade milk, growth & fertility in one generation.



Disease control

No bull movement between herds = fewer venereal & transboundary diseases.



Low cost & safety

No dangerous bull to keep & feed; a straw costs a fraction of a bull.



Records & traceability

Known sire = pedigree, genomic selection & planned matings.



Carabao upgrading

AI with **Murrah / river-type** semen is the engine that converts a low-yield swamp buffalo herd into a productive dairy herd — village by village, straw by straw.

This is precisely the PCC mission at national scale.

Semen: collection → freezing → storage

1

Collection

Artificial vagina or electro-ejaculation from proven sires.

2

Evaluation

Motility, concentration & morphology checked immediately.

3

Extension

Diluted in egg-yolk / soy extender with cryoprotectant.

4

Filling & freezing

Into 0.25 / 0.5 mL straws; cooled, then frozen in LN₂ vapour.

5

Storage

Held in liquid nitrogen at **−196 °C** — viable for decades.

Each straw is identity-labelled: bull, breed & batch — the foundation of traceable breeding.

Straw handling: the details that decide conception

- ✓ Work fast in the tank neck — lift the canister **no higher than the frost line**, ≤10 s exposure
- ✓ Thaw in a water bath at **35-37 °C for 30-45 s** — never in a pocket or by air
- ✓ Dry the straw, load a **pre-warmed** gun, protect from cold shock & sunlight
- ✓ Inseminate **within ~15 minutes** of thawing
- ✓ One straw thawed at a time

35-37°C

thaw temperature · 30-45 s

-196°C

tank storage · avoid re-cooling a thawed straw

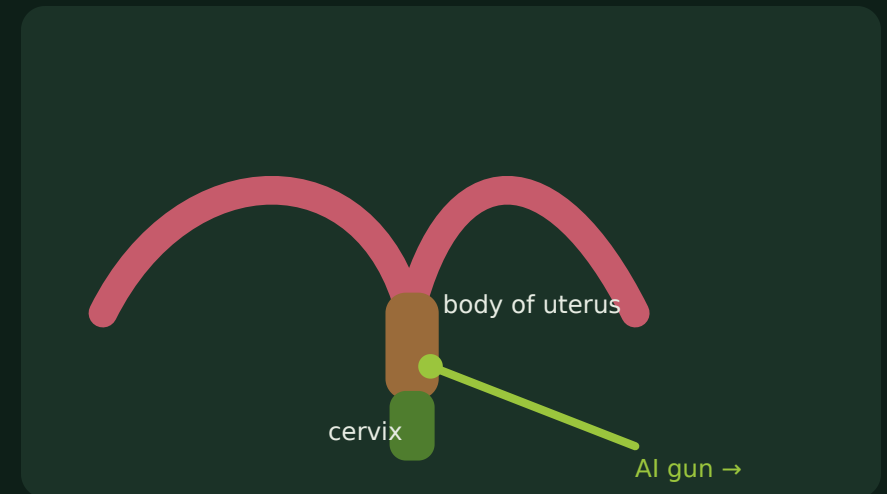
Cold shock & slow thawing kill sperm — good semen fails from poor handling.

The recto-vaginal technique

- Gloved hand in the **rectum** grips the cervix; the gun passes through the vagina
- Manipulate the cervix onto the gun — thread through its rings
- Deposit semen in the **body of the uterus** (site AI), just past the cervix
- Deposit slowly; withdraw gently; minimize stress & trauma

Golden rule

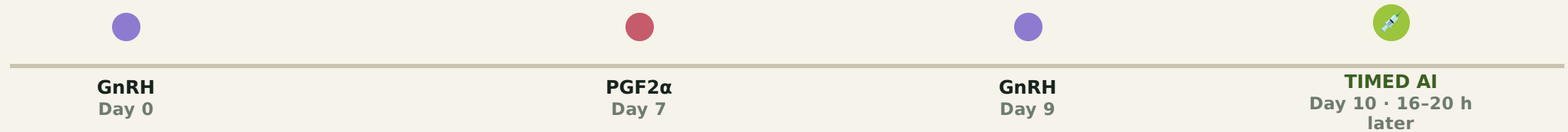
"Quick, clean, gentle." A calm animal & clean pass protect the pregnancy you are trying to create.



Deposition site: body of the uterus (not the vagina, not deep in a horn).

Ovsynch → timed AI: breeding by the calendar

Control ovulation itself, and you can inseminate every animal at a set hour —
no heat detection at all.



Why it transforms buffalo breeding

Silent & seasonal heat no longer matters — the whole group is bred on schedule, lifting herd pregnancy rate.

Add progesterone (CIDR)

For non-cycling / anestrus animals, a CIDR insert during the protocol jump-starts cyclicity.

Why a "good" AI still fails — the checklist

Heat / timing

Wrong timing or missed silent heat — the #1 cause.

Semen handling

Poor thawing, cold shock, delay after loading.

Technician skill

Deposition site & gentleness — training pays back fast.

Nutrition & BCS

Thin or over-fat animals don't conceive — energy balance matters.

Health & uterus

Retained placenta, metritis, lameness lower fertility.

Heat stress

Shade, water & cooling — critical for buffalo in the tropics.

Fertility is a system. Fix the weakest link — measure, don't guess (that is what Part 5 is about).

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PART FOUR

Embryo Transfer & IVF

If AI multiplies the *sire*, embryo transfer multiplies the **elite female** — turning one exceptional donor into a whole generation.

Why embryo transfer (MOET & IVF)?

A cow normally gives **~1 calf a year**. With embryo technology, one elite donor can produce **dozens of calves a year** — carried by ordinary recipient females.

- A** **MOET** — superovulate the donor, AI, flush embryos ~day 7
- B** **OPU-IVF** — aspirate eggs from live donors, fertilize in the lab
- C** Transfer fresh, or **freeze / vitrify** for later & for export



For scarce elite buffalo

Superior river-buffalo females are **rare & valuable**. ET/IVF is the fastest way to propagate them nationally — and to conserve endangered genetics.

Honest note: buffalo superovulate less predictably than cattle — OPU-IVF is often the better route.

Superovulation & donor management

1

Select & sync donor

Proven, healthy, cycling elite female; synchronize the wave.

2

FSH stimulation

Decreasing FSH doses over ~4 days → many follicles grow at once.

3

PGF2 α & heat

Luteolysis → heat; multiple follicles ready to ovulate.

4

Insemination

AI with elite sire (often 2×) to fertilize many eggs.

Typical yield: a handful of transferable embryos per flush in cattle — variable, and lower in buffalo.

Embryo recovery & grading

- **Day 7 non-surgical flush** — a catheter in the uterine horn, warm media collects the embryos
- Search under the microscope; identify stage: **morula** → **blastocyst**
- Grade **1 (excellent)** → **4 (degenerate)** on shape, colour & cell compactness
- Only good embryos are transferred or frozen



Morula



Blastocyst



Expanded

Grade drives outcome

Grade-1 fresh embryos give the best pregnancy rates; grade-1 & 2 freeze well.

OPU–IVF: making embryos in the laboratory

- 1 OPU** — ultrasound-guided egg aspiration from live donors (repeatable, even pregnant/young)
- 2 IVM** — oocyte maturation in culture (~24 h)
- 3 IVF** — fertilize with (sexed) semen
- 4 IVC** — culture ~7 days to blastocyst
- 5 Transfer** fresh, or **vitrify** at -196°C

Advantage for buffalo: OPU sidesteps their poor superovulation response.

OUR IVF EMBRYOS — BLASTOCYST STAGE



Sexed semen & sexed embryos

To grow a **dairy or breeding herd**, you want females. Flow-sorted **sexed semen (~90% female)** and sexed IVF embryos let you plan the sex of the calf.

- Faster herd expansion — more replacement heifers
- More heifers born to your best females via IVF
- Fewer unwanted male calves in a dairy system

~90%
female calves with sexed semen



Buffalo dairy

Sexed genetics accelerates building a river-buffalo **dairy** herd from a limited elite base.

From frozen embryo to a live calf

- 1 **Synchronize recipients** — their cycle must match the embryo's age
- 2 **Check the CL** — transfer only to females with a good corpus luteum
- 3 **Thaw / warm** the embryo & load the transfer gun
- 4 **Transfer** non-surgically into the uterine horn on the CL side
- 5 **Pregnancy check** ~30 days by ultrasound

Elite genetics, carried by an ordinary female — **this is how a nation upgrades a herd in years, not decades.**



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PART FIVE

Smart Farming for Reproduction

Sensors and data turn reproduction from guesswork into measurement — and directly solve the buffalo **silent-heat** problem.

RUMEN BOLUS SENSOR, INSIDE THE ANIMAL



SMART FARM · SENSOR-BASED DETECTION

A sensor that never sleeps — so it catches silent heat

A rumen bolus continuously measures **inner body temperature (± 0.01 °C)**, rumination & activity — reading estrus and health from inside the animal, 24 hours a day.

🔥 Heat alerts, day or night

Activity + temperature flag estrus even when signs are invisible — the buffalo breakthrough.

📅 Insemination window

Tells you the ideal AI hour per animal — precision timing.

🐮 Calving prediction

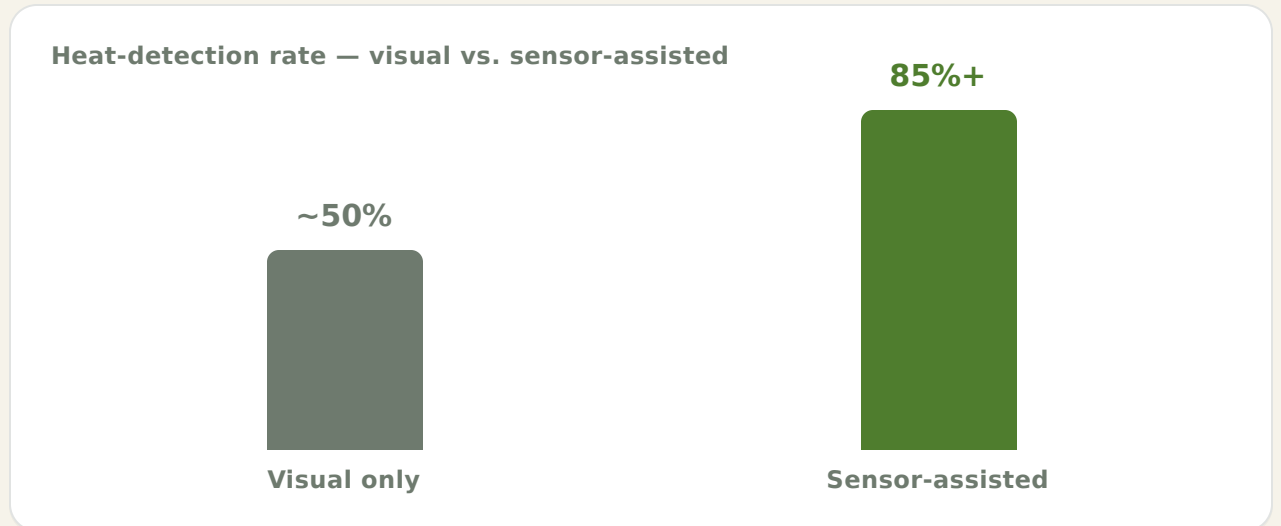
Alerts ~15-30 h before calving — fewer losses.

❤️ Early illness

Fever & metritis flagged days early — protects fertility.

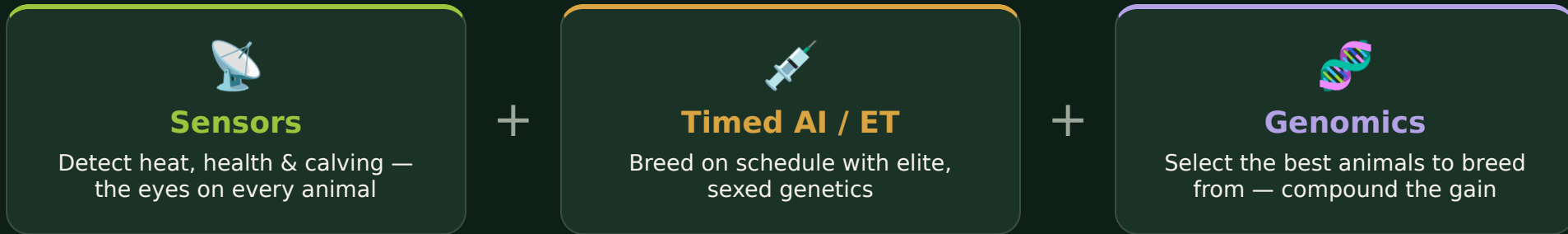
From alerts to action — the reproduction dashboard

- ✓ Automatic **heat & AI work-lists** each morning
- ✓ **Protocol scheduling** — Ovsynch / CIDR steps flagged on the right day
- ✓ **Pregnancy & open-day tracking** per animal & per group
- ✓ Herd **KPIs** trend over time — decisions from data, not memory



Illustrative — sensors recover heats that eyes miss, especially at night. Higher detection → higher pregnancy rate.

Precision reproduction: three tools, one system



For the carabao program: sensors solve silent heat → timed-AI & ET spread superior genetics → genomics picks the next donors. A national herd, upgraded on data.

The reproduction KPIs to manage by

CR

Conception rate

% pregnant per AI

S/C

Services / conception

lower is better ($\approx 1.5-2$)

CI

Calving interval

target $\sim 12-14$ mo (longer in buffalo)

PR

Pregnancy rate

detection $\times$ conception

You cannot improve what you do not measure. Track these per animal, per group, per season — and act on the trend.

Five things to remember

1

Reproduction is the **engine of genetic gain** — master it and you upgrade the whole herd.

2

Missed heat is the #1 failure. For buffalo, plan for silent & seasonal heat.

3

Timed AI & synchronization beat the silent-heat problem — breed by the calendar.

4

ET & IVF multiply elite females — the fast track for scarce buffalo genetics.

5

Sensors & data make it precise, repeatable & scalable nationwide.

Practical order for a farm: detect heat well → synchronize → AI on time → measure → add ET & sensors.

Thank you — Salamat po

Questions, discussion & hands-on sessions welcome. I look forward to supporting the carabao improvement mission through reproduction & smart-farm technology.

Hyunjae Ha, DVM

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AI

Embryo Transfer

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